Annotazioni di `main_revision.pdf`

testo marcato e note — tutti i tipi

Totale annotazioni: 223 — 72 sottolineate (70 con nota), 125 evidenziate (123 con nota), 26 barrate (4 con nota). Note totali: 197.
Ordine: per pagina e posizione, come nel PDF originale. Ogni voce indica il tipo di marcatura. Un raro simbolo matematico nel testo marcato non estraibile è reso come [?]; le note sono integrali.

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1. [Sottolineato] Pagina 2
Testo: esis characterizes the transport statistics of real soarin and compares them with stochastic transport models. nt state of the project. A first, reproducible acquisitio
Nota: It more about developing a good mode which is able to reproduce data statistics. It's not just a comparison. Even if the subject is not developed at this stage, the thesis will also analyze the motion of the gliders during a competition, trying to notice what are the key points which differs from solo flights.

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2. [Sottolineato] Pagina 2
Testo: ument r 212,602 glider f
Nota: Is it right?

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3. [Sottolineato] Pagina 2
Testo: has colle 203,343 (2001–2
Nota: Is it right?

---

4. [Sottolineato] Pagina 2
Testo: nce ac 9,259 g and
Nota: Is it right?

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5. [Sottolineato] Pagina 2
Testo: 999–2000 192,971 ion is o
Nota: right?

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6. [Sottolineato] Pagina 2
Testo: hang-gl 192,768 d to fur
Nota: right?

7. [Barrato] Pagina 2
Testo: PS track may be Contest
Nota: will be (for sure)

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8. [Evidenziato] Pagina 2
Testo: s .
Nota: The paragraph starting with "A first, reproducible acquisition....." and ending with "sing the same approach": I think it has been written here in the abstract even if it doesn't really fit it. I mean: it's more about a description of the raw dataset and it should be placed in the right section in the core of the thesis. It's too specific to be in an abstract. An abstract should be a very short summary of the thesis content and purpouse.

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9. [Sottolineato] Pagina 5
Testo: ng birds carry no continuous s from the moving atmosphere sunlight heats the ground. A cr
Nota: I don't like "the "moving atmosphere" as an expression

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10. [Barrato] Pagina 5
Testo: ating cycle of t energy-e"cient a search phase
Nota: along a nearly straight path toward ..

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11. [Sottolineato] Pagina 5
Testo: is the setting in on of a flight, me
Nota: is one of the... (Obviously, there are other ways to develop anomalous transport)

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12. [Evidenziato] Pagina 5
Testo: ↑ ↓ t with ω ↔= 1, as opposed to the linear growth (ω = 1) of Continuous-time random walks and Lévy walks are the standard 1,2]; anomalous transport has been reported in animal movement, ordinary Brownian di!usion. Con frameworks for such regimes [1,2]; in disordered media, and in turb
Nota: It's not like that. First of all pure CTRW doesn't allow to work with angular memory/persistence. To do so I need a persistence random walk. Second, LW is a sub-caregory of CTRW, it's a coupled CTRW. What likely is going to be a good model here is a sort of mix between a persistent random walk and a CTRW where between a ballistic leg and the next there is a waiting time and the direction between the legs is correlated.

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13. [Evidenziato] Pagina 5
Testo: inuous-time random walks and Lévy walks are the standard anomalous transport has been

reported in animal movement, ulent flows. Whether the trajectories of real soaring flights frameworks for such regimes [1,2]; anomalous in disordered media, and in turbulent flows belong to an anomalous regime, and of whic
Nota: Citations missing. When citing where anomalous transport has been detected, I need to support the statement citing some papers

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14. [Sottolineato] Pagina 5
Testo: and approximately between di!usive ed analysis based independent of aircraft type (H = 1/2 and ballistic H = 1 on the transition–search–clim
Nota: I like this clarification. If it can help, when H lies between 0.5 an 1, the motion is characterized as superdiffusive.

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15. [Sottolineato] Pagina 5
Testo: on the transition–search–climb cycle. This thesis builds on it: it en re-examines both the phase segmentation and the transport analysis.
Nota: The thesis also revisits the cleaning and filtering process, a step at the very veginning of the analysis

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16. [Evidenziato] Pagina 5
Testo: three objectives:
Nota: Actually, it persues other goals as well, like: - defining a correct segmentation strategy; - analyze phase motion statistics; - gathered, filter and ... the competiton dataset and see how competion modify key motion properties.

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17. [Evidenziato] Pagina 6
Testo: and to compare the empirical statistics with stochastic to test for anomal transport models.
Nota: As already detailed in a previous note, it's not just about comparing empirical statistics with existing transport models but it's more about developing the right minimal model which is able to reproduce empirical key findings.

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18. [Evidenziato] Pagina 6
Testo: ata collection is ongoing and may incorporate further The second and third objectives define the work ahead sources, nota (Chapter 3).
Nota: Firstly, there should be more than only three objectives, as stated in a note before. Secondly, data cleaning/filtering is a working topic. What's ahead the cleaning/filtering is what define future tasks

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19. [Sottolineato] Pagina 6
Testo: ified and is describ notably sailplanes r 3).

Nota: Other datasets will be added, especially the ones concering sailplanes and competition flights

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20. [Sottolineato] Pagina 6
Testo: This document reports the current state of the project and is revised as the work proceeds Its quantitative content—the summary statistics and tables—is generated directly from the dataset, so it remains consistent with the data in hand. Chapter 2 follows the data end to end: Its quantitative content—the summary statistics and dataset, so it remains consistent with the data in hand the acquisition and the reproducibility of its pipeline
Nota: A reader could confuse theiself while reading this sentence. It seems like only summary statistics and table are data driven. It obvious that every plot,table come form data? Where else would it come from, then?

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21. [Evidenziato] Pagina 6
Testo: ables—is generated directly from the Chapter 2 follows the data end to end: Sec. 2.1), the dataset and its statistics, dataset, so it remains consistent with the data in hand. Chapter 2 follows the data end to end: the acquisition and the reproducibility of its pipeline (Sec. 2.1), the dataset and its statistics, and the pre-processing that turns the raw logs into analysis-ready trajectories. Chapter 3 sets the acquisition and the reproducibility of its pipeline (Sec. 2.1), the dataset a and the pre-processing that turns the raw logs into analysis-ready trajectories out the planned transport analysis and modelling.
Nota: This description of what Chapter 2 is about it's quite a mess. It doen't cite the rights sections (apart from Sec. 2.1) and I fear it's outdated too. You should carefully read Chapter 2 and rewrite this brief summary.

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22. [Barrato] Pagina 6
Testo: he raw logs into and modelling ps to the scien
Nota: —

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23. [Barrato] Pagina 7
Testo: the French free-f on separate sites). Pilots
Nota: —

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24. [Evidenziato] Pagina 7
Testo: one for hang gliders (pilot, date, take-o! he raw GPS tracklog (delta.ffvl.fr). Pilots upload their flights; each e and landing sites, scored distance, duration, glider) in IGC format (the standard flight-recorder format,
Nota: Are these onyl some of the actual metadata that each flight bring with it? I think so but I'm not sure. If this is only a patial list of the available metadat, I recommend you to list all the available metaddata or be clear about the partiality of the list.

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25. [Sottolineato] Pagina 7
Testo: ost cases, the raw GPS tracklog Both sites expose identical URL cquisition pipeline handles both. in IGC format (the standard flight-recorder format, Sec. 2.2). Both sites expose identical URL structure and machine-readable XML exports, so the same acquisition pipeline handles both. A "season" is identified by its starting year, e.g. the year 1999 denotes the 1999–2000 season;
Nota: I think this clarification is more appropriate to place it in the section "Implementation details"

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26. [Evidenziato] Pagina 7
Testo: catalog is built (catalog.csv, one row per flight, metadata plus the together with a per-season index (). The raw data From these, a derived local path of its track) (tens of gigabytes) live
Nota: I want a better description of waht catalog actually contains. It has many columns, which are metadata I suppose. I want the document very precise on what are the metadata gathers in the catalog.csv file

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27. [Evidenziato] Pagina 7
Testo: data only (tens of gigabytes) live on an external the small per-season index is versioned
Nota: Does make sense to commit the catalog.csv too?

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28. [Evidenziato] Pagina 8
Testo: me (implementation details: Sec. 2.1.3). Both FFVL all 186,052 paraglider and 6,716 hang-glider downloaded IGC standard (Sec. 2.2). collections have been verified in full: all 186,052 par tracklogs pass validation against the IGC standard
Nota: The numbers reported here differ from what has been reported in the abstact. Please carefully check what are the right ones

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29. [Evidenziato] Pagina 8
Testo: y source of the present study, but 203,343 flights, 186,225 tracklogs ghts, 6746 tracklogs) are collect
Nota: One again, the number differ form those reported elsewhere.

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30. [Evidenziato] Pagina 8
Testo: er data (203,343 flights, 186,22 (9,259 flights, 6,746 tracklogs) n of Sec. 2.4. Additional public
Nota: The numbers still differ from those reported elsewhere

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31. [Sottolineato] Pagina 8
Testo: araglider a XContest alog.
Nota: Clarify that even the Xcontest data has been requested but, as of today, July 20th, 2026, netither WeGlide nor XContest have replied to my requests

---

32. [Sottolineato] Pagina 8
Testo: gh further portals such as XCo merged into the same catalog.
Nota: What do you mean? Catalog.csv are different depending on the dataset/category.

---

33. [Barrato] Pagina 8
Testo: f the , not 2.1), recorders used in free inferred from the files, so conformance is che
Nota: —

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34. [Barrato] Pagina 9
Testo: The validity flag is fixed by the rather than the horizontal position altitude), whereas marks a two-d
Nota: —

---

35. [Evidenziato] Pagina 9
Testo: andard to two values and certifies the GNSS altitude A marks a valid three-dimensional fix (usable GNSS mensional fix or a GNSS drop-out, in which case the rather than the horizontal position: A marks a valid three-dimensional altitude), whereas V marks a two-dimensional fix or a GNSS drop-out, standard requires the GNSS-altitude field to be written as zero. A fix ca
Nota: I don't like the way you have expressed the meaning of A/V. In fact, even if the GNSS receiver fails to resolve the altitude, the flight keeps being a 3d flight thanks to the baro altitude. On the other hand, looking at what you've written, it seems like every time the GPS altitude fails the flight become 2d.

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36. [Sottolineato] Pagina 9
Testo: and quantified in Sec. 2.7.1; th drives the phase segmentation. ce of records is thus a time
Nota: I'm not sureabou "to drive" is the best verb to use here. In fact, the vertical velocity will be one one the variable that will play a roje in detecting the three phases.

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37. [Sottolineato] Pagina 9
Testo: data as record attributes, which the fields used downstream are: tes and the French department;
Nota: It's not about what will be used and what won't. It' about listing all the metadata available

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38. [Evidenziato] Pagina 9
Testo: o! and landing sites and the Fre (aile) and a class (aile_class, ype (the scoring category: free di
Nota: aile or wing? What does raw xlm data report?

39. **[Sottolineato]** **Pagina 9**
Testo: ublic pag versioned g a track,
Nota: Should be versioned catalog.csv too?

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40. **[Evidenziato]** **Pagina 10**
Testo: rigid- glide paragliders (flexib framed delta wing; ratio25). They d
Nota: Are delta wing always rigid framed? What dataset metadata tells us? Looking at the dataset and at what has been wirttend in the next paragraphs it seems like paragliders could have also non rigid wings, nemaly flexible wings.

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41. **[Barrato]** **Pagina 10**
Testo: rigid aircraft; fastest, glide – which is why the reference ratio ↭25). They di!er several-fold in cruise speed and turn r study [4] reports its transport statistics separately for each. In practice the category is set by the data source, since t
Nota: The reason why vilpellet et al report them separately is not only because they show huge differences in cruise speed, glide ratio, turning-radius and other key metrics. In fact, is way more meaningful to analyze them separately being different "airplanes"

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42. **[Sottolineato]** **Pagina 10**
Testo: lysis is limited to the two FFVL disciplines. Within each discipline the flights can be further stratified by performance class, recorded ; the two disciplines use di!erent class systems. FFVL labels paragliders on the Within each discipline the flights can be further stratified by performance class, recorded in aile_class; the two disciplines use di!erent class systems. FFVL labels paragliders on the EN/AFNOR certification ladder – EN A ("A ou 1", most stable, recreational), EN B ("B ou in aile_class; the two disciplines use di!erent class systems. FFVL labels paragliders on the EN/AFNOR certification ladder – EN A ("A ou 1", most stable, recreational), EN B ("B ou 1-2"), EN C ("C ou 2"), EN D ("D ou 2-3"), the "CIVL Competition Class" (CCC) racing EN/AFNOR certification ladder – EN A ("A ou 1", most stable, recreational), EN B ("B ou 1-2"), EN C ("C ou 2"), EN D ("D ou 2-3"), the "CIVL Competition Class" (CCC) racing wings, plus tandem ("biplace") and non-certified ("non homologuée") entries. Hang gliders 1-2"), EN C ("C ou 2"), EN D ("D ou 2-3"), the "CIVL Competition Class" (CCC) racing wings, plus tandem ("biplace") and non-certified ("non homologuée") entries. Hang gliders follow the FAI/CIVL class system, which in this data reduces to flexible wings ("Delta Class wings, plus tandem ("biplace") and non-certified ("non homologuée") entries. Hang gliders follow the FAI/CIVL class system, which in this data reduces to flexible wings ("Delta Class 1" and the intermediate "Delta Class Sport") and rigid wings ("Rigide Class 2" and "Rigide follow the FAI/CIVL class system, which in this data reduces to flexible wings ("Delta Class 1" and the intermediate "Delta Class Sport") and rigid wings ("Rigide Class 2" and "Rigide Class 5"). In both cases the ladder tracks performance (higher classes fly faster and flatter), 1" and the Class 5"). so it is the
Nota: A table here is a better way t drive the aile classification for both para and hang.

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43. **[Evidenziato]** **Pagina 10**

Testo: must be normalised or placeholder class.
Nota: It's unclear what "normalization" means here and what it involves.

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44. [Sottolineato] Pagina 10
Testo: ibe the FFVL CFD dataset currently in hand; further sources may It comprises 212,602 flights in total, of which 192,971 carry a GPS been downloaded and verified, split across two disciplines. be added later (Sec. 2.1). It comprises 212,602 flights in total, of which 192,971 carry tracklog and 192,768 have been downloaded and verified, split across two disciplines. The paraglider collection spans 27 seasons, from 1999–2000 to 2025–2026: 203343
Nota: Please, as already noted before, check if this number are correct or if they are made-up numbers.

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45. [Sottolineato] Pagina 10
Testo: cklog and 192,768 have been downloaded and verified, split across two disciplines. The paraglider collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights, which 186225 (91.6 %) carry a tracklog and 186052 (99.9 %) are downloaded and verified The paraglider collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights, of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 2.1). The hang-glider collection spans 25 seasons, from 2001–2002 to 2025–2026: 9259 of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 2.1). The hang-glider collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6746 (72.9 %) carry a tracklog and 6716 (99.6 %) are downloaded and verified (Table 2.1). The hang-glider collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6,746 (72.9 %) carry a tracklog and 6,716 (99.6 %) are downloaded and verified (Table 2.2). The hang-glider archive is roughly twenty times smaller, reflecting the smaller flights, of wh (Table 2.2). active popu
Nota: —

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46. [Evidenziato] Pagina 12
Testo: hout a trac no IGC file
Nota: Saying "no IGC file" means the same thing as "Without GPS" in tables 2.1 and 2.2?

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47. [Sottolineato] Pagina 12
Testo: A small residue of unrecoverable files server does not actually serve as a valid IG
Nota: Do you mean the file with the GPS track whose download failed?

---

48. [Evidenziato] Pagina 12
Testo: accuracy e any en
Nota: cosa sarebbe ""accuracy" qui? cosa intendi? mi sembra sensa senso

---

49. [Evidenziato] Pagina 12
Testo: cal dynamics over a quarter to the GNSS (Sec. 2.7.1), b of paraglider altitude, flag
Nota: almost a third

---

50. [Evidenziato] Pagina 12
Testo: dates. Som but flagged
Nota: Writing "flagged" here doen't really help the reader to understand what is the action behind? What does "flagged" mean in practice?

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51. [Sottolineato] Pagina 12
Testo: to a sequence of fixes (ti, εi, ϑi, hi) – UTC time, geodetic latitude and by the soaring.analysis.igc parser, which reads the B records at ions and returns both altitude channels. The pipeline then proceeds longitude, and altitude – by the soaring.analysis.igc parser, their fixed character positions and returns both altitude channels in a fixed order:
Nota: That has more to do with "implementaion details" section.

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52. [Barrato] Pagina 13
Testo: kage. The ordering is chosen so that each step runs in the frame where it is exact and cheap. ps (i)–(iv) act on the raw geographic coordinates, since the only geometric quantities they
Nota: –

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53. [Evidenziato] Pagina 13
Testo:) to the metric East–North–Up with the origin at the take-o! data and referred to the right
Nota: !!! Beacuse of the trimming, the origin can't be at the take off

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54. [Evidenziato] Pagina 13
Testo:) = 0), so the tangent-plane frame is built from good data and referred to the right The final steps (vi)–(vii), which produce filtered position, velocity and acceleration, are ly metric and so run after the conversion. origin. The final steps (vi)–(vii), which produce fil naturally metric and so run after the conversion.
Nota: I don't like how the reason here reported that tries to explain why the final steps have been performed after the ENU conversion. I don't like it becaude one can ocjects that even the velocity threshold was a metric quantity but we have applied it in the phase ii) and iv) even if we hadn't already applied the ENU conversion.

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55. [Barrato] Pagina 13
Testo: – vertical velocity, climb/sink, and the phase segmentation that rests om the barometric altitude rather than the GNSS altitude. The reason The vertic on them – is one of si
Nota: –

56. [Sottolineato] Pagina 13
Testo: is one of signal quality in the freq has sub-metre relative precision constant o!set from the ICAO
Nota: Does it really have sub-metre precision?

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57. [Sottolineato] Pagina 13
Testo: etric ; its er 10
Nota: Full stop and capital letter indicating the start of a new sentence

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58. [Sottolineato] Pagina 13
Testo: rs are slow: a (read directly g the synoptic constant o!s in Fig. 2.1a) pressure and
Nota: Actually, to be precise, The figure 2.1a shows a gap between the baro and the gps altitude. Saying that the gap is due to the baro offset means stating that the gps is very precise in term of offset.

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59. [Sottolineato] Pagina 13
Testo: m the ICAO reference, tens of metres up to order 100 m (read directly a drift of tens of metres over a multi-hour flight, following the synoptic tmosphere's departure from the ICAO profile (Fig. 2.1b). Slow errors are in Fig. 2.1a), and a drift of tens of metres over a multi-hour flight, follow pressure and the atmosphere's departure from the ICAO profile (Fig. 2.1b) annihilated by the di!erencing that every dynamical quantity involves (a ve
Nota: The drift showed in fig 2.1b is linked to a flight of half an hour, more or less. So it's inappropriate to refer to fig 2.1b while talking about a multi-hour flight.

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60. [Sottolineato] Pagina 13
Testo: tical velocity, a is geometrically 5–2 times worse displaceme referenced than the h
Nota: It's not crystial clear what "geometrically referenced" mean

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61. [Sottolineato] Pagina 13
Testo: orrelation). The GNSS altitude, the vertical dilution of precision ath adds high-frequency jitter o
Nota: I don't like this way of expressing the concept

---

62. [Sottolineato] Pagina 13
Testo: noisier: the multipath plifies. Thi
Nota: It's not obvious what multipath is.

---

63. [Sottolineato] Pagina 13
Testo: ree, tracking the real cl though not uniformly. mmon floor set by the m
Nota: Unclear what this sentence means

---

64. [Sottolineato] Pagina 13
Testo: di!er, though not the common floor o the median spect
Nota: what do you exactly mean when talking about “”the floor”? Do you mean the ending part of the plot 2.1C where a constant PSD is retrieved?

---

65. [Sottolineato] Pagina 13
Testo: cal dilution of precision, which amplifies exactly e poor GNSS reception, while mos that band the unifor
Nota: One should briefly explaining why high frequency noise is amplified when a finite difference is computed. Please keep it brief.

---

66. [Sottolineato] Pagina 14
Testo: entative flight over a 30 min window: the o!set between the pressure and o!set also drifts by tens of metres the two raw channels ellipsoidal references over the flight, so on
Nota: I see what you mean, but I think a reader might find it a little hard to follow.

---

67. [Sottolineato] Pagina 14
Testo: ellipsoidal refe over the flight, spectrum of ea
Nota: Actually fig 2.1b shows that the offset drifts by ten of metres in 30 mins. Then In a multi-hour flight the offsets could drift more

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68. [Evidenziato] Pagina 14
Testo: Source recorded. Which channel a flight uses is stored as a per-flight field – an alt_source column (or) in the processed dataset, alongside duration, fix count and the ce recorded. Which channel a flight uses is stored as a per-flight field – an alt_source column (baro or gnss) in the processed dataset, alongside duration, fix count and the like; the raw IGC files are never modified. column (baro or gnss) in the processed d like; the raw IGC files are never modified.
Nota: I think the best plave for this paragraph is in the “details implementation”. question: when spekaing about fix count, do you mean the rw numer of fixex each IGC file have, tight? beacuse the cleaning is performed after the altituide channel coiche. one more thing: adding a column where the entry is always the saem (baro or gnss) isn’t very efficient in terms of occupied space.

---

69. [Evidenziato] Pagina 14
Testo: the few ed from ability. A flight is used with its barometric individual fixes with a missing barometric value GNSS. A logger with no pressure sensor write
Nota: How do many flights with a working baro altituide have fixes missing baro altituide? how

many fixes do miss the baro altituide (min_number, mean_number, max_number)? Removing those fixes would make a gap in the trajectory? I mean, if those fixes are concetuives removing them will produce a gap. Is that common? Does it ever happen?

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70. **[Sottolineato]** **Pagina 14**
Testo: paraglider flights scanned carry no barometric channel (implementation details: (The counts quote the census scan directly, through generated macros; the archive
Nota: I don’t understand netiher the meaning and the porpouse of this sentence nor if it should be palce here instead of writing it in the computation details section ath the end of the document. Moreover, it should be challenged if it is really usefull or if it can be just deleted without causing any demage.

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71. **[Sottolineato]** **Pagina 15**
Testo: keeps growing by a trickle, so they can trail the live download totals of Sec. 2.5 slightly until the scan is refreshed.) keeps growing by a tri the scan is refreshed.)
Nota: I don’t understand netiher the meaning and the porpouse of this sentence nor if it should be palce here instead of writing it in the computation details section ath the end of the document. Moreover, it should be challenged if it is really usefull or if it can be just deleted without causing any demage.

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72. **[Sottolineato]** **Pagina 15**
Testo: ng and flight-lev geodetic angles t-circle (haversi
Nota: geodetic coordinates

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73. **[Sottolineato]** **Pagina 15**
Testo: the geodetic angles (εi, ϑi). Dista great-circle (haversine) distances utive fixes,
Nota: Implementation/Python REF: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.pairv

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74. **[Sottolineato]** **Pagina 15**
Testo: L = [?] i !si The spherical is ↫03 %, far and the extent D = maxi ! approximation is ample here below the GPS noise), and
Nota: rephrase this sentence

---

75. **[Sottolineato]** **Pagina 15**
Testo: cal far een approximation is amp below the GPS noise cleaned.
Nota: Comparing the ellipsoidal correction with the gpn noise require that you declere what the gpd noise really is.

---

76. [Sottolineato] Pagina 15
Testo: [?] (fix0, fixi), the farthest a fix gets from take-o!. The spher the ellipsoidal correction to an inter-fix distance is ↫0.3 % avoids converting coordinates before the trajectory has b
Nota: This sentence is a bit vague. Please specify exactly what the cited “ellipsoid correction” really means and where the 0.3% comes from.

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77. [Sottolineato] Pagina 15
Testo: ack (at most the paraglider speed bound of Table and typically within the outlier floor ϖmin = 15 m genuine fix removed from a slow, horizontally locali
Nota: It doen’s work. You’re citing epsilon_min but it hasn’t been definite yet.

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78. [Evidenziato] Pagina 15
Testo: and back within one step – is its residual rk – the great-circle
Nota: —

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79. [Evidenziato] Pagina 16
Testo: distance of the fix from the component-wise median of the fixes within ±w of tk – is scaled by the robust local spread of those residuals, and the fix is flagged when distance of the fix from the component-wis the robust local spread of those residuals,
Nota: It’s better to explain the Hamper identifier using math and formulas instead of using words. Using math should make the concept easier to be understood.

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80. [Sottolineato] Pagina 16
Testo: y – bound that rules the step physically impo flagged, yet flying at an ordinary speed – A fix on the trend agrees with its neigh
Nota: Using dashes here is a bit AI style. I don’t like it too much

---

81. [Evidenziato] Pagina 16
Testo: o, so it escapes the ou A value out of range, t needs no context.
Nota: It seems like only issues with the altitude falls in this cathegory. If so, please be clearer in explaining that. I mean, It not just a value out of range but it’s a value realted with the z-axis. Am I wrong?

---

82. [Evidenziato] Pagina 16
Testo: t needs (Status: boun
Nota: I would avoid writing the status in this way, at the end of the paragraph. I think it’s better to report these this information along tiht the main text. Apart from the ending sentence: “The routine that walks a trajectory applying the three detectors is still to be built”. It’s good where it has been placed beacuse it’s a status annotation.

83. [Evidenziato] Pagina 16
Testo: und (1 ↘100, and fro
Nota: Why neagtive altitude are allowed? Does it make sense?

---

84. [Evidenziato] Pagina 16
Testo: nd audite ςz, φfreeze utine tha
Nota: These two params hasn't been declared yet

---

85. [Evidenziato] Pagina 16
Testo: the Hampel and frozen-lock parameters (k, w, to be fixed by the injected-defect calibration. he three detectors is still to be built.)
Nota: mmmh, I think this is an overshoot. I mean, we should fix this params and then, a posteriori, see if the number of removed fixes is too low/high. i'm suggesting having this approcach beacuse it's a bit lighter and it requires less time to be implemented. what do you think? Of course we can procede with "he Hampel and frozen-lock parameters (k, w, ϖmin, ϖ, ςz , φfreeze) are a priori working values, to be fixed by the injected-defect calibration." if you think it's really the best and if you think it's worth it.

---

86. [Evidenziato] Pagina 16
Testo: allest whose removal restores the trend. The action then depends on the channel a!ected, under one rule: the corrupt value is moved, and its reconstruction is deferred to the single resampling step (Sec. 2.7.6); nothing The action then depends on the channel a!ected, under one rule: the corrupt value is removed, and its reconstruction is deferred to the single resampling step (Sec. 2.7.6); nothing is imputed in place. A corrupt altitude keeps position and time and marks only the altitude removed, and its reconstruction is deferred to the single resampling step (Sec. 2.7.6); nothing is imputed in place. A corrupt altitude keeps position and time and marks only the altitude as missing. The fix still samples the path, and the missing value is restored at resampling, is imputed in place. A corrupt altitude keeps position and time and marks only the altitude as missing. The fix still samples the path, and the missing value is restored at resampling, so the segmentation receives a complete () within each retained segment. A corrupt as missing. The fix still samples the path, and the missing value is restored at resampling, so the segmentation receives a complete (x, y, z) within each retained segment. A corrupt horizontal position or time instead deletes the fix: what remains without them, a timestamp so the segmentation receives a complete (x, y, z) within each retained segment. A corrupt horizontal position or time instead deletes the fix: what remains without them, a timestamp and an altitude, no longer samples the path. The deletion is what makes the defect count as horizontal position or time instead deletes the fix: what remains without them, a timestamp and an altitude, no longer samples the path. The deletion is what makes the defect count as a hole. Either way the horizontal values at that time must be reconstructed; the question and an altitude, no longer samples the path. The deletion is what makes the defect count as a hole. Either way the horizontal values at that time must be reconstructed; the question is whether the resampling step sees a hole there. Seeing it, that step applies its gap rule: a hole. Either way the horizontal values at that time must be reconstructed; the question is whether the resampling step sees a hole there. Seeing it, that step applies its gap rule: bridge a short hole, split at a long one. Had the corrupt nodes been kept instead, with the is whether the

resampling step sees a hole there. Seeing it, that step applies its gap rule: bridge a short hole, split at a long one. Had the corrupt nodes been kept instead, with the position merely marked missing, ten consecutive corrupt fixes would read as ten samples of a bridge a short hole, split at a long one. Had the corrupt nodes been kept instead, with the position merely marked missing, ten consecutive corrupt fixes would read as ten samples of a continuously recorded stretch; the gap rule would never fire, and the resampler would lay an position merely marked missing, ten consecutive corrupt fixes would read as ten samples of a continuously recorded stretch; the gap rule would never fire, and the resampler would lay an invented straight line across a stretch that may hold a full thermalling circle. The barometric continuously recorded stretch; the gap rule would never fire, and the resampler would lay an invented straight line across a stretch that may hold a full thermalling circle. The barometric sample lost with a deleted node costs little: that channel is smooth enough to bridge with invented straight line across a stretch that may hold a full thermalling circle. The barometric sample lost with a deleted node costs little: that channel is smooth enough to bridge with error below the sensor noise (Sec. 2.7.1). sample lost with a deleted node costs li error below the sensor noise (Sec. 2.7.1). A defect that no bounded removal rep
Nota: Tutta questa parte è davvero poco chiara. It's poorly explained. A better way to convey the message is required. si deve capire be eil perchè si applica un trattamento diveros a problemi sulla Z e su XY. Inoltre non si capisce appunto perchè, dato che problemi sula z vegnono interpolati amntenendo il fix, non si possa fare lo stesso con i problemi su x,y o sul tempo. certo, mi va anche bene otliere il fix, però deve essere chiaro il perchè.

---

87. [Sottolineato] Pagina 16
Testo: value is nothing altitude removed, and its rec is imputed in place. as missing. The fix
Nota: I don't like this sentece. re-phrase it please

---

88. [Evidenziato] Pagina 16
Testo: or below the sensor noise (Sec. 2.7.1). A defect that no bounded removal repairs – a position discontinuity, such as a re-acquisition et, where the track jumps and stays – is not a deletion problem: both sides are genuine A defect that no bounded removal repairs – a position discontinuity, such as a re-acquisition o!set, where the track jumps and stays – is not a deletion problem: both sides are genuine and only the transition between them is unknown, so the trajectory is split at the step, like a o!set, where the track jumps and stays – is not a deletion problem: both sides are genuine and only the transition between them is unknown, so the trajectory is split at the step, like a long gap (Sec. 2.7.6, which states the boundary-censoring rule). and only the transition between them is unknown, so the traject long gap (Sec. 2.7.6, which states the boundary-censoring rule).
Nota: Why does this clarification has been written in the section "Removal policy"? It has more to do with splitting a trajectory in two or more segments. And should be explained really well what this means, what effects this has, and when one proceeds cutting the trajectory.

---

89. [Evidenziato] Pagina 17
Testo: Spatial diameter tion: the run's b
Nota: It has not been specified how long is the time window used to compute the diameter!

---

90. [Barrato] Pagina 17

Testo: , not step. bounding d
Nota: —

---

91. [Evidenziato] Pagina 17
Testo: -circle distance bet working value 15 m gh currently set eq
Nota: It' s really dangerous to repor in the text a working value. In fact, if it will change, it will be harder to detect all the values written down in the text. then, it coulb be wiser to refer to a table where this working values are reported. Rurthermore, this 15m should be the value for \{}epsilon right? it's not the value for the GPS noise. right? If so, specify that.

---

92. [Evidenziato] Pagina 17
Testo: by at least a The criterion vances10 m factor of two (quantif is the run's diameter, per fix, comparable t
Nota: It has not been specified how long is the time window used to compute the diameter!

---

93. [Barrato] Pagina 17
Testo: ed below), and clears the test on the diameter alone. The criterion never the fix-to-fix step: at 1 Hz a climbing wing advances ↓10 m ϖ, so a step-based test would flag every climb. is the run's diameter, never the fix-to-fix step: at 1 Hz a climbing win per fix, comparable to ϖ, so a step-based test would flag every climb.
Nota: —

---

94. [Sottolineato] Pagina 17
Testo: ling keeps g over the run A horizontall
Nota: be more precise when saying "over the run". It's not clear what the run is

---

95. [Evidenziato] Pagina 17
Testo: itude is still and only one n (iii).
Nota: Are we sure when saying "only one"?

---

96. [Sottolineato] Pagina 17
Testo: y same bits the zeroed en horizont
Nota: what do you mean saying "zeros"?

---

97. [Sottolineato] Pagina 17
Testo: split The bing
Nota: New line here

98. [Evidenziato] Pagina 18
Testo: ails the cut on tw R = v2/(g tan ↽) 30 m, and hang
Nota: Where does this formula come from? Write a brief footnote to exaplin its provenance and the ration behind

---

99. [Sottolineato] Pagina 18
Testo: once. Geometrically, a circling win v ↗10 m s↔1 at a bank of ↽↗35→ wider (2R ↭50 m), so against ϖ =
Nota: Are these values realistic?

---

100. [Evidenziato] Pagina 18
Testo: wh 7 r
Nota: why 7? It seems 3.3 to me

---

101. [Evidenziato] Pagina 18
Testo: climb of order 1 m s↔ sustained over φ a flatness tolerance ςz of a few metres. timescale: is about two thermallin
Nota: It seems to me that the the numerical value of the flat tolerance hasn't been deciearede yet. So, how and why it has been written ""few meters"?

---

102. [Evidenziato] Pagina 18
Testo: tim so di
Nota: new line here

---

103. [Evidenziato] Pagina 18
Testo: pans at least two full circles of any genuine climb, whose ike the other detector parameters it is a working value, to ibration. Cutting a genuine climb on a barometric flight diameter test (i) then sees in full; like the o be fixed by the injected-defect calibration. therefore requires both margins to fail at on
Nota: are we really sure we want to follow this procedure on the params calibration? It seems like an expensive procedure and, maybe, a simpler approach woudl be beneficial in this case. I mean, we could try to set this values and then try to assess their reasonableness a posteriori.

---

104. [Evidenziato] Pagina 18
Testo: sp ; a
Nota: fullstop

---

105. [Evidenziato] Pagina 18
Testo: njected-defect c both margins writes.
Nota: I dont like this "both margin". I get the meaning but for a first-time reader it could be not so clear what it means.

---

106. [Evidenziato] Pagina 18
Testo: The rule assumes a functioning barometric channel and weakens t – the GNSS altitude cannot stand in as the witness, because it Residual corner cases. The rule assumes a functioning barometric channel and weakens where that channel is absent – the GNSS altitude cannot stand in as the witness, because it comes from the very receiver whose lock is in doubt and freezes with the position it would be where that channel is absent – the GNSS altitude cannot stand in as the witness, because it comes from the very receiver whose lock is in doubt and freezes with the position it would be checking. On the GNSS-fallback minority (Sec. 2.7.1) the recorder's declarations of (iii) – the comes fro checking. flag or
Nota: What it's written here is in open contrast with the previous paragraph. In fact, we have allowed the usage of the GNSS altitude every time this was the onluy one available. And the procedure described above uses the gnss altitude. I'm confused.

---

107. [Evidenziato] Pagina 18
Testo: Residual corner cases. where that channel is abs
Nota: Apart from all the annotations made, this paragraph is, in general, very difficult to follow. The reader get confused reading the section beacuse the misisng case are neither typographically nor logically well separeted and well written. Maybe, but I'm not sure, a dotted list may help.

---

108. [Barrato] Pagina 18
Testo: ld be – the her a checking. On the GNSS-fallback minority (Sec. 2.7.1) the r V flag or zeroed GNSS altitude, or a byte-identical repeat collapsed run is cut, and both residual error cases are con
Nota: —

---

109. [Evidenziato] Pagina 18
Testo: ons of alone - circ
Nota: saying alone may be misleading since they are also present the diameter condition and the duration condition. Am I wrong?

---

110. [Sottolineato] Pagina 18
Testo: are confined there: a sub-ϖ circling climb jittering freeze – repeated positions that not byte-identical – is kept wrongly; each carrying a V flag is cut wrongly, and an undeclared jittering freeze – di!er by a metre or two of receiver noise, so they are not byte-identical is a near-empty set, and the default to keep favours preserving real
Nota: Please wirte a better explaination of what "jittering freeze" really is.

---

111. [Evidenziato] Pagina 18
Testo: that each nign di!er by a metre or t is a near-empty set, miss remains on bar
Nota: I think we cannot say that a priori. It could be said only after checking real data.

---

112. [Evidenziato] Pagina 18
Testo: bounded cost, since w unrecorded wind drift
Nota: What is that???

---

113. [Evidenziato] Pagina 18
Testo: Figure 2.2. out-of-range over a large
Nota: Not just the out-of-range but also the off-the-trend

---

114. [Barrato] Pagina 18
Testo: so the sparse tail the bounds act on is visible. Panel (a) hang gliders carry a non-decaying plateau of implausible), and the bound sits where the real decay stops. Binning shows why the speed bound is per-discipline: hang sustained ground speeds (55 m s^{-1} to 90 m s^{-1}), and choices, sample sizes and the validation of the rob
Nota: —

---

115. [Sottolineato] Pagina 18
Testo: real decay stops. Binning (implementation details: choices, s Sec. 2.6.2
Nota: Qui forse fa citato i lparagrafo dedicato in implmentation details, ovvero "Diagnostic sample (Figure 2.2). "

---

116. [Evidenziato] Pagina 19
Testo: usable altitude and the altitude The asymmetry with a corrupt ter of information rather than drops out as missing. Neither case needs handling of its own. The asymmetry with a corrupt horizontal position (fix deleted outright, Table 2.3) is a matter of information rather than severity: without its position a node is no longer a point of the path, and deleting it keeps horizontal position (fix deleted outright, Table 2.3) is a matter of information rather than severity: without its position a node is no longer a point of the path, and deleting it keeps the hole visible to the gap accounting (removal policy above), whereas a fix lacking only the severity: without its position a node is no longer a point of the path, and deleting it keeps the hole visible to the gap accounting (removal policy above), whereas a fix lacking only the altitude still carries the horizontal sample. the hole visible to the gap accounting (rem altitude still carries the horizontal sample.
Nota: Are we sure this is the right palce for this clarification? I think it's not. Maybe a beter place, before of after, could be found. Furthermore, I'm not sure we need this paragraph "The IGC validity flag". If so, it should be explained better. If not, its content should be place somewhere else, inside other sections.

---

117. [Evidenziato] Pagina 19
Testo: he vertical vel delete a node on a GNSS-fa
Nota: Do you mean delete a fix?

---

118. [Evidenziato] Pagina 19
Testo: ack flight) – because within each retained lled by interpolation the guar segment per chan
Nota: For a segment, do you mean the trajectory between a gap and the next?

---

119. [Evidenziato] Pagina 19
Testo: , the short holes filled by interpolation The invariant is thus “complete (x, y, z)
Nota: I don’t understand this sentence

---

120. [Evidenziato] Pagina 20
Testo: within a segment”, not across a split. rule it feeds, need no repair of their o
Nota: I don’t understand this sentence!

---

121. [Evidenziato] Pagina 20
Testo: hy the validity flag, and the missing-altitude a dropped altitude is not a lost altitude but e a single, audited interpolation does all the rule it feeds, need no repair of their own here: a dropped altitude is not a lost altitude but a deferred one, reconstructed downstream where a single, audited interpolation does all the filling. a defer filling.
Nota: ??? questo succede sempre?? Non proprio immagino. Cioè, se per tanti secondi non ho l’altituidine interpolo lo stesso? mi sembra pericoloso.

---

122. [Barrato] Pagina 20
Testo: ght’s airborne fixes, the flight is no longer – the cleaning counterpart of the sampling- being cleaned but rebuilt, regularity cut (Sec. 2.7.6).
Nota: —

---

123. [Evidenziato] Pagina 20
Testo: Validating the cleaning. shows the cleaning is not too
Nota: Mi sembra una bellisisma strategia. Questo quindi vuol dire che tutti i parametri del fix cleaning vanno settati in questo modo? se si va bene ma voglio questa sezione più dettagliata e precisa, elencando esattemtne la procedura da eseguire.

---

124. [Evidenziato] Pagina 20
Testo: That each bound removes only a negligible fraction of fixes ggressive; it says nothing about what the cleaning misses. Two

Nota: This statement is not supported by data. What I mean is that it seems to me that we haven't checked if the fix-level cleaning actually removes only a small fractions being not too agressive or, on th eother hand, it removes many fixes

---

125. [Evidenziato] Pagina 20
Testo: ches of known size are planted in clean track measures the false-negative rate directly. – the displacement-scaling exponent and th
Nota: Do you mean the number of corrupted fixes that have not been detected?

---

126. [Barrato] Pagina 20
Testo: measures the false-negative rate directly. Invariance of the – the displacement-scaling exponent and the waiting-time tail g is tightened, and the operating point is taken on the plateau r – w
Nota: —

---

127. [Evidenziato] Pagina 20
Testo: 2.7.3 Trimming of ground phases
Nota: It's better to make two paragraph inside this section, one for the outer gound phase and one for the mid-flight landing and relauch pahses

---

128. [Evidenziato] Pagina 20
Testo: ustained-speed rule read forward and time-reversed – not a cut at the first which a slow soaring phase (ground speed near zero when hovering into wind) e trigger mid-flight. The persistence prevents a gust or a GPS glitch on sustained drop, which a slow soaring would otherwise trigger mid-flight. the ground from being mistaken for
Nota: This sentence it's not really clear in its meaning

---

129. [Evidenziato] Pagina 20
Testo: he stint is kept. A cut inheriting the segment nts shorter than stint is treated exactly like a long gap: excised and split (Sec. 2.7 rules – boundary censoring, no second pass of the flight-level cuts. are not cut but flagged: a wait with sustained and a flat
Nota: This sentence is not clear

---

130. [Evidenziato] Pagina 20
Testo: d pass of t sustained () fits
Nota: What do you mean by "sustained"? For how long?

---

131. [Evidenziato] Pagina 21

Testo: The the as a sensitivity check – the same flag-without-deleting pattern as the outlier identifier. The parameters (Tground and the flatness tolerance) join trimming in the configuration when the guard is built (implementation details: Sec. 2.6.3). parameters (Tground and the flatness tolerance) joi guard is built (implementation details: Sec. 2.6.3). One caveat is structural: the criterion is a ground
Nota: frase non chiara

---

132. [Evidenziato] Pagina 21
Testo: – the same flag-without-deleting pattern as the outlier identifier. nd the flatness tolerance) join in the configuration when
Nota: frase non chiara

---

133. [Evidenziato] Pagina 21
Testo: hase adjacent to take-o! the trimmed fraction is old and the persistence or landing (a launch straight into ridge lift, say) can recorded per flight so this failure mode is auditable. a!ect what fraction of each trajectory is retained, so
Nota: it's usefull to build and report an histogram of the trimmed fraction frecorded and deleted per flight. Two histograms if there is any difference between a recorded and a deleted.

---

134. [Evidenziato] Pagina 21
Testo: le. Both the threshold an so they are made explicit
Nota: It's important to explain how this threshold are set

---

135. [Sottolineato] Pagina 21
Testo: flights and inflates the minimal-threshold that still clears non- apparent p strategy: genuine flig
Nota: bold or italics?

---

136. [Evidenziato] Pagina 21
Testo: ersistence of the MSD. The flight-level filtering therefore uses a minimal-threshold for each criterion, the threshold is set to the smallest value that still clears non- hts (sled runs, tows, aborted or truncated logs), so the ensemble is disturbed as little strategy: for each criterion, the threshold is set to the genuine flights (sled runs, tows, aborted or truncated logs) as possible. Once chosen, each cut is verified by the full
Nota: ok a parole, ma nella pratica come si fa? come si capisce quando una threshold è minimale??

---

137. [Barrato] Pagina 21
Testo: nges almost nothing. All ; the thresholds follow the e a flight, plus a metadata flight-level

quantities are computed directly minimal-threshold strategy just described. stratifier:
Nota: —

---

138. [Evidenziato] Pagina 21
Testo: d attempts. An upper sanity bound T ⇑16 h removes the census holds fifteen “flights” of 16 h to 166 h, with ld poison the long-lag MSD. The cut is deliberately loggers left running past the flight: the census holds fifteen “flight paths up to 10,000 km, which would poison the long-lag MSD. decoupled from the maximum MSD lag – the ensemble at lag a
Nota: Are we sure of these numbers? The code of this census should be in the repo. It should not be only a disposable code use only once but something that can be consulted. ATTENZONE, ATTENZIONE, ATTENZIONE: il fatto che ci siano dei controlli sual dataset come questo e che di sicuro sia stato runnato un codice va bene ,MA, questo codcie deve essere nelal repo. Ongi volta che viene suato del codice per fare qualcosa che finsice in questo domcuento il codice deve essere nella repo e ci deve essere il rifeirmento al codice, magari il rifeirmento si mette nella seizone implementation details.

---

139. [Evidenziato] Pagina 21
Testo: L ⇐10 km, erate logs. A al-threshold
Nota: This threshold should be fixed at 20 km

---

140. [Barrato] Pagina 21
Testo: , with L = [?] i !si from the great-circle steps of (2.1), a An earlier draft used 30 km; the census showed that value d strategy itself. Above 40 min, 10 km catches 6 flights [?] floor against degenerate logs. An earlier draft violating the minimal-threshold strategy itself. out of 184583, while 30 km removed 522
Nota: —

---

141. [Barrato] Pagina 21
Testo: used 30 km; the census showed that value Above 40 min, L < 10 km catches 6 flights genuine, localized soaring flights (median violating the minimal-threshold strategy itself. Above 40 min, L < 10 km catches 6 flights out of 184,583, while L < 30 km removed 5,229 genuine, localized soaring flights (median extent 74 km) – precisely the most-trapped tail of the ensemble, whose exclusion would out of 184,583, while L < 30 km removed 5,229 genuine, localized soaring flights (median extent 7.4 km) – precisely the most-trapped tail of the ensemble, whose exclusion would bias the transport toward travellers. Duration already clears tows and sled runs. extent 7.4 km) – precisely the most-trapped tail of the ensemble, whose exclusion bias the transport toward travellers. Duration already clears tows and sled runs.
Nota: —

---

142. [Evidenziato] Pagina 21
Testo: vehicle log, or a dead sens retained paraglider records;
Nota: Does the census hold any candidate among hang-gliders?

---

143. [Evidenziato] Pagina 21
Testo: cannot be b-analyses Glider class (metadata). For class-strati normalised onto the discipline's system but kept in the pooled one. Stratificat
Nota: e quali sarebbero?

---

144. [Barrato] Pagina 21
Testo: A separate minimum-fix-count cut would be redundant: at T ⇐40 min even a 10 s logger – the slowest common hang-glider rate (Fig. 2.7) – already records ↓240 fixes, ample for the A separate minimum-fix-count cut would be redundant: at T ⇐40 min even a 10 s logger – the slowest common hang-glider rate (Fig. 2.7) – already records ↓240 fixes, ample for the kinematics; the thin tail of still slower loggers is handled by the !t stratification (Sec. 2.7.6), the slowest common hang-glider rate (Fig. 2.7) – already records ↓240 fixes, ample for the kinematics; the thin tail of still slower loggers is handled by the !t stratification (Sec. 2.7.6), not by a fix-count cut. A further criterion, sampling regularity, is applied later, where a flight's kinematics; the thin tail of still slower loggers is handled by the !t stratification (Sec. 2.7.6), not by a fix-count cut. A further criterion, sampling regularity, is applied later, where a flight's time base is examined (Sec. 2.7.6) (implementation details: Sec. 2.6.4). not by a fix-count cut. A further criterion, sampling regularity, is applied time base is examined (Sec. 2.7.6) (implementation details: Sec. 2.6.4).
Nota: –

---

145. [Evidenziato] Pagina 22
Testo: Figure 2.3. discipline (p
Nota: poichè volgio che la thershold vegna spostata da 10km a 20km e questo richeided (forse) ri-runnare su tutto il datastee, che è una rpcoedrua lunga e pesante, direi che conviene salavrsi i dati che vanno a comporre l'istorgammi e i plots sotto, in modo tale che se devo cambiare nuovament ela thershold sarà necessario ri-runnare solo i plots e gli historgammi invece che rirunnare su tutto il dataset.

---

146. [Barrato] Pagina 22
Testo: n the full dataset – every parsed track, per – not from the declared metadata and not tion and (b) flown path length. Bottom: the discipline (pa sub-sampled. fraction of fli
Nota: –

---

147. [Evidenziato] Pagina 22
Testo: ts retained versus (c) th marginal, not cascaded) ut in the tail. (implemen
Nota: non chiaro cosa significa

---

148. [Evidenziato] Pagina 22
Testo: raphic to local Carte are naturally metric ed surface and canno
Nota: questo naturally metric non è chiarisismo. Fa sembrare gli altri step come del tutto non

metrici, ma abbiamo usato la haversine distance diverse volte. sii più chiaro qui

---

149. [Barrato] Pagina 22
Testo: e and cannot be di!erenced following the convention of directly to obt Vilpellet et al.
Nota: —

---

150. [Evidenziato] Pagina 23
Testo: torial plane and the ellipsoid normal at the point (Figure 2 which is the angle to the line from the Earth's centre O. cause the normal does not pass through (Figure 2.4):
Nota: frase scritta male

---

151. [Evidenziato] Pagina 23
Testo: ellipsoid the two di!er, because the norm ε ↘←reaches ↗0.19→(about 11↑) near 45 fix by up to ↓20 km.
Nota: IS this estimate right?

---

152. [Barrato] Pagina 23
Testo: umes it. The longitude ϑ carries since the ellipsoid is a surface of longitude are exactly the () no such ambiguity: it is the same revolution about the polar axis. that orient the local frame below
Nota: —

---

153. [Sottolineato] Pagina 23
Testo: revolution about the polar axis. that orient the local frame below.
Nota: a quale immagine ci si riferisce? va citata

---

154. [Evidenziato] Pagina 23
Testo: the take-o! isplacement Ste fix,
Nota: è il fix di take off oppure e comuqnue abbastanza succesivo al take off il primo fix che sopravive al trimming?

---

155. [Evidenziato] Pagina 24
Testo: (2.6)
Nota: questa formula è davvero corretta? sicuri al 100%? crea un'appedice in cui mostri i passagig che portano a questa formula. In realtà, nella stessa appendice, prima metti i passaggi che portano alla fomrula 2.5 in cui c'è la converione da WGS84 a ECEF. insomma fai un appendice su questi due steps e sui dttagli mateamtic dietro queste formule, che altriemnti semrbano un po" cascate dal cielo. In generale, per le appendici, se questa viene citata per la prima volta nel capitoo 2

allora sarà 2.(Lettera maiuscorla), e.g.: 2.C. quindi se un'appendice è citata per la prima votla nel chapther 3 allora la prima sarà 3.A, ovvero Appendix 3.A

---

156. [Evidenziato] Pagina 24
Testo: sian frame: its horizontal plane (E, N) is the "north–east" so there are not two distinct coordinate systems but one (global) ECEF representation. Over the spatial extent of a
Nota: frase davvero poco chiara.

---

157. [Evidenziato] Pagina 24
Testo: (global) ECEF representation. Over the spatial extent of a the tangent-plane approximation is accurate to well below the action of the rotation matrix: it turns a vector's global single flight (te the GPS noise. ECEF compon
Nota: non si capisce, a livello quantitativo, come questa frase è supportata. ok dirlo ma va fatto un rifeirmento quantitativo. non si capisce bene cosa vuol dire altrimenti.

---

158. [Sottolineato] Pagina 24
Testo: ns of kilometres) the tangent-plane approximation is accurate to well below Figure 2.5 shows the action of the rotation matrix: it turns a vector's global nts into local East, North, Up ones. the GPS noise. Figure 2.5 shows the action of the r ECEF components into local East, North, Up ones.
Nota: metti questa frase nella caption della figura a cui si riferisce e togli questa frase da qui.

---

159. [Evidenziato] Pagina 24
Testo: The altitude h that enters this transform is the adopted barometric channel (Sec. 2.7.1); changing channel would move (E, N) by a relative h/(N + h) ↓10↔5, i.e. negligibly, so the horizontal coordinates are insensitive to the altitude source. The vertical coordinate z = U changing channel would move (E, N) by a relative h/(N + horizontal coordinates are insensitive to the altitude source is therefore, up to the small deterministic tangent-plane curv
Nota: non mi è chiaro per nulla. Non mi è chiaro da dove viene la formula h/(N+h) e in che senso cambinado il channel di altitude cambierebbe il rifeirmento EN. nella pratica su cosa avverrebbe questo cambio? sii più preciso. ora è fumoso

---

160. [Evidenziato] Pagina 24
Testo: nistic tange z ↗h ↘h0
Nota: super attenzone qui. Nei paragrafi di prima non è mai stato detto che l'altitudine viene posta a zero nel punto di take off. Volendo possiamo fare questa cosa ma deve essere dichiarato prima. anche se sinceramente non lo farei. non vedo perchp andare a perdere dell'informazione. in questo modo si avrebbe un piano tangente che però non tocca terra ma viene sollevato no?

---

161. [Sottolineato] Pagina 24

Testo: time base is exa 69 % of flights), d (Fig. 2.7) – bu
Nota: c'è il codcie che porta a scirvere quesot numero? il codice è nella repo?

---

162. [Barrato] Pagina 24
Testo: drop-outs. We keep each flight : a common grid would either he resolution of fast ones, and at its native rate rather than resampling to a common cadence: a common grid would either upsample slow loggers (inventing information) or throw away the resolution of fast ones, and
Nota: —

---

163. [Sottolineato] Pagina 25
Testo: it is unnecessary, since the derivative operator below is given the flight's own !t (Sec. 2.7.7). What matters is uniformity within a flight, not across flights.
Nota: —

---

164. [Evidenziato] Pagina 25
Testo: Uniform
Nota: se non ho gap ma ho un numero di fixes mancanti superiore al 10%, il volo viene scartato? Se si, in quale sezione/paragraof è staat scritta questa cosa? è staat scritta in modo chairo? fa forse messa nella sezione in cui si parla di flight-level cleaning. perchè è un criterioc he prota allelimianzione di tutto un flight. però magari quel flight ha più segments. e magari solo alcuni segments vengono buttati. quindi questo controllo fa fatto dopo che il volo è stato splittarto? e questo dove avviene?

---

165. [Evidenziato] Pagina 25
Testo: ly irregular. Isolated jitter or sin gmax = min [?]10 !t, max(20 s, 2 !t) [?] thresholds, Sec. 2.6.6): the flight is interpolating across the small gaps o
Nota: mettere rifeirmentoa al paragrafo " The split bound has two scales." in cui viene spegata l'origine di questa threshold

---

166. [Sottolineato] Pagina 25
Testo: circle, and the segmentation downstream would read the invented straight The absolute cap of 20 s is pinned by three constraints: at least 2⇓the dence (10 s, Fig. 2.7), so a single missed fix never splits a flight; at most stretch as a glide. The absolute cap of 20 s is pinned by three constraints: at least 2⇓the slowest common cadence (10 s, Fig. 2.7), so a single missed fix never splits a flight; at most about two thirds of a thermalling period, so a bridged gap can never hide a full circle; and well slowest common cadence (10 s, Fig. 2.7), so a single missed fix never splits a flight; at most about two thirds of a thermalling period, so a bridged gap can never hide a full circle; and well below = 60 s, for consistency – an excised frozen-lock run of that length is split, so a about two thirds of a thermalling period, so a bridged gap can never hide a full circle; and well below φfreeze = 60 s, for consistency – an excised frozen-lock run of that length is split, so a native gap of comparable length cannot be bridged. One refinement keeps the first constraint below φfreeze = 60 s, for consistency – an excised fr native gap of comparable length cannot be

bridged. true at every cadence: the cap is floored at 2 !, i.e
Nota: conviene fare un elenco puntato per maggior chiarezza?

---

167. [Evidenziato] Pagina 25
Testo: ds (gmax = 10 s); from !t = 2 s to 10 s the 20 s cap does; beyond, the 2 !t floor. On the census, the combined bound leaves 15.3 % of paraglider and 40.5 % of hang-glider hts with at least one split, against 11.8 %/14.5 % under the relative rule alone; the increase On the census, the combined bound leaves 15.3 % of paraglider and 40.5 % of hang-glider flights with at least one split, against 11.8 %/14.5 % under the relative rule alone; the increase sits at the slow cadences, where a 20 s hole is only a few missed fixes. A split is cheap for flights with at least one split, against 11.8 %/14.5 % under the relative ru sits at the slow cadences, where a 20 s hole is only a few missed fixes. the ensemble observables (below) but fragments the within-segment st
Nota: deve essre presente il codice di questo census che ha protato a questi numeri. il codice sempre nella repo. non si capisce bene chi è la combined bound e la realtive rule.

---

168. [Evidenziato] Pagina 25
Testo: cheap for the cap's Sec. 2.7.2 the ensemble observables (below) but fragments the within-segment statistics, so the cap's working value is one of the parameters swept in the downstream-invariance check of Sec. 2.7.2 before being frozen. working value is on before being frozen
Nota: non capisco quale value deve essere fissato sinceramente. cos'è questo cap's working value?

---

169. [Evidenziato] Pagina 25
Testo: 20 s hole is only a few missed fixes. A spli but fragments the within-segment statistics ters swept in the downstream-invariance check
Nota: sinceramente non credo che studierò le within segment statistics. sarò interessato alle statistiche globali e a quelle per fase. ok forse tu stai dciendo che segmentando troppo creo un effetto sulle statistiche delle tre fasi. se è così allora sii più chiaro qui nel drie gli effetti di un eccessiva segmentazione.

---

170. [Evidenziato] Pagina 25
Testo: elapsed time referred to Ensemble quantities (the not the path between: a the parent take-o!, and the split only assigns a segment identifier. Ensemble quantities (the MSD, the propagator) need the position at a given elapsed time, not the path between: a segment keeps its parent's origin and clock (Sec. 2.7.8), and a GPS position is absolute, so the MSD, the propagator) need the position at a given elapsed time, not the path between: a segment keeps its parent's origin and clock (Sec. 2.7.8), and a GPS position is absolute, so the positions after the gap remain valid and the flight simply contributes no data at the missing segment keeps its parent's origin and clock (Sec. 2.7.8), and a GPS position is absolute, so the positions after the gap remain valid and the flight simply contributes no data at the missing lags. Path-dependent quantities (path length, phase durations, step lengths, time-averaged posi lags wind
Nota: molto importnate, metti in rislato questa frase/concetto

---

171. [Sottolineato] Pagina 25
Testo: at the missing time-averaged utions follow. lags. Pa windows
Nota: che intendi?

---

172. [Barrato] Pagina 26
Testo: handled with censoring-aware estimators, ery split would read as a spuriously short as censored and excluded from th rather than counted as complete; wait and thin the same tail the c
Nota: Sarei per non usare censoring-aware estimators. Se il numero di durations flagged and discarded non è eccessivo allora semplicemente andrei a costruire le phi(t) non contando le fasi che hanno un gap in mezzo.

---

173. [Sottolineato] Pagina 26
Testo: from the ↼(φ) fits mplete; otherwise
Nota: sono censored anche nella costruzione degli istogrammi di \{}phi(t)?

---

174. [Sottolineato] Pagina 26
Testo: eraged MSD stratified by duration (Ch. 3), a segment beginning at is an observation window of an aged process, and is used as such d. parent time t0 > 0 is rather than re-zeroed.
Nota: frase non chiara. spiega meglio

---

175. [Evidenziato] Pagina 26
Testo: No second pass of the flight-level cuts. The parent flight has already passed the cuts of Sec. 2.7.4, and re-applying the duration cut to segments would preferentially discard the No second pass of the flight-level cuts. The parent flight has already passed the cuts of Sec. 2.7.4, and re-applying the duration cut to segments would preferentially discard the data behind late-flight gaps. A segment needs only to support the smoothing window Sec. 2.7.4, and re-applying the duration cut to segments would preferentially discard the data behind late-flight gaps. A segment needs only to support the smoothing window and the within-segment statistics: one below a minimal duration (a configuration key data behind late-flight gaps. A segment needs only to support the smoothing window and the within-segment statistics: one below a minimal duration (a configuration key set with the implementation, Sec. 2.6.6) is dropped alone, and the rest of the flight is and the within-segment statistics: one below a minimal duration (a configuration key set with the implementation, Sec. 2.6.6) is dropped alone, and the rest of the flight is retained. set with retained.
Nota: questo punto non l'ho capito. è spoegato davvero male. e davverr non ci ho capito nulla di questo punto. o si spiga meglio, o, se non serve, si toglie

---

176. [Sottolineato] Pagina 26
Testo: The parent flight has already passed the cuts of cut to segments would preferentially discard the No second pass of the flight-level cuts. The parent flight has already passed the cuts of Sec. 2.7.4, and re-applying the duration cut to segments would preferentially discard the data behind late-flight gaps. A segment needs only to support the smoothing window Sec. 2.7.4, and

re-applying th data behind late-flight gaps. and the within-segment stat
Nota: frase non chiara

---

177. [Sottolineato] Pagina 26
Testo: per channel asures direct
Nota: cosa intendi con per channel? x,y,z o ppure z_baro e z_gnss? chi è il channel qui? maggiore chiarezza qui

---

178. [Evidenziato] Pagina 26
Testo: d. The shape- has left horizontal position, which the transport preservingly (a monotone piecewise-cubic) behind; the barometric altitude, a smoot
Nota: spuega meglio cos'è e perchè usaimo questa

---

179. [Sottolineato] Pagina 26
Testo: analysis measures directly, is interpolated shape- and only ever across the short gaps a split has left h, low-noise, slowly-drifting sensor within a phase preservin behind; (Sec. 2.7.
Nota: non mi è chiaro per nulla. uno split pensavo che dividesse il volo in più segments. invece mi sembra di capire che viene fatta un'interpolazione!! ma in che senso? e poi questa inteprolazione viene fatta separatamente per x ed y o vien considerta una funzione che considera insieme x ed y, ovvero un funaizone vettoriale?

---

180. [Barrato] Pagina 26
Testo: itude, a sm linearly (or e complete
Nota: —

---

181. [Sottolineato] Pagina 26
Testo: owly-drifting sensor within a phase with negligible error over such gaps. thin a retained segment every grid
Nota: qiesta frase va supportata da osservazioni quantitative, altrimenti non ha senso.

---

182. [Evidenziato] Pagina 26
Testo: ownstream is the invariant m reconstruct
Nota: non si capisce chi è questo invariante. devi essere più chiaro per favore!!

---

183. [Evidenziato] Pagina 26
Testo: easured from reconstructed s inherits the same caution .7.2 (a dropped barometric
Nota: si ma nella pratica non ci cpaisce cosa comporta questa caution

184. **[Sottolineato]** **Pagina 26**
Testo: a bridg deferred restored
Nota: deferred è il temrine giusto da usare qui?

---

185. **[Evidenziato]** **Pagina 26**
Testo: SS-fallback flight) are restored here. One might expect the altitude to tolerate wider gaps, since the vertical speed is about an er of magnitude smaller than the horizontal (1 m s1 against10 m s1). It does One might expect the altitude to tolerate wider gaps, since the vertical speed is about an order of magnitude smaller than the horizontal (vz ↓1 m s↔1 against vxy ↓10 m s↔1). It does not, and the gap bound is the same for every channel: the interpolation error is set by the order of magnitude smaller than the horizontal (vz ↓1 m s↔ against vxy ↓10 m s↔). It does not, and the gap bound is the same for every channel: the interpolation error is set by the curvature of the signal over the gap, not by its speed (a fast but straight motion is interpolated not, and the gap bound is the same for every channel: the interpolation error is set by the curvature of the signal over the gap, not by its speed (a fast but straight motion is interpolated exactly, a slow but turning one is not), so a slower channel is not automatically a smoother curvature of the signal over the gap, not by its speed (a fast but straight motion is interpolated exactly, a slow but turning one is not), so a slower channel is not automatically a smoother one. exac one. M
Nota: siamo sicuri che questo discros debba essere fatto qui? secondo me andava fatto un po" più sopra quando si parlava della grandezza del gap massimo accettabile oltre il quale avveine uno split

---

186. **[Evidenziato]** **Pagina 26**
Testo: –glide transition a thermal entry terpolation. One falling inside it; and those transitions are exactly whe flips vz from about ↘1.5 to +3 m s↔1 in a second or two) asymmetry is real and already used: an isolated missi
Nota: siamo davvero sicuri di qeusti valori? da dove vengono?

---

187. **[Sottolineato]** **Pagina 26**
Testo: ause the barometric channel is so smooth. T but it says nothing about widening the gap. sampling-regularity control anticipated in
Nota: non capisco questa frase

---

188. **[Sottolineato]** **Pagina 26**
Testo: tude only" at cleaning, but it says nothing about widening the gap. The gap bound is the sampling-regularity control anticipated in Sec. 2.7.4; kes the fraction of flights it leaves whole auditable in the same way as the du
Nota: make this sentence clearer

---

189. **[Evidenziato]** **Pagina 26**

Testo: in the same way as the duration and One caveat carries downstream: a few !erence velocity, a turning angle), so path-length cuts (implementation details: Sec. 2.6.6). One caveat carries downstream: a few observables are intrinsically !t-dependent (a finite-di!erence velocity, a turning angle), so where they are compared across flights the ensemble is stratified by rate or restricted to the observables are intrinsically !t-dependent (a finite-di!erence velocity, a turning angle), so where they are compared across flights the ensemble is stratified by rate or restricted to the dominant one, rather than pretending a single cadence. where they are compared across flights the ensemble is dominant one, rather than pretending a single cadence.
Nota: è davvero necessario applicare una stratification quando studio quantià come la velcoità o il turn angle? non posso semplicmente prendere tuti i voli e costruire le statisitche di queste varibaili? non vedo che problema ci potrebbe/dovrebbe essere usando voli con delt_t diverso.

---

190. [Evidenziato] Pagina 27
Testo: Figure 2.6 (every parse
Nota: anche qui conviene salvarsi i dati che vanno a creare questi plots in mod tale che se questi pot devono esere ricreati cambianod threshold o colori o altre cose grafiche, allor alo si possa fare senza stare a ri-runnare su tutto il dataset.

---

191. [Evidenziato] Pagina 27
Testo: Sampling-regularity diagnostics, flight, not only those passing th
Nota: forse titolo non adeguato. mi sembra non descriva bene chi sonoq eisti 4 plots

---

192. [Evidenziato] Pagina 27
Testo: ap, relative to the flight's native over the bulk of the distribution; adopted cut (dashed) sits on th
Nota: in che senso? PERCHè?

---

193. [Evidenziato] Pagina 27
Testo: interval, over the bulk of the distribution; integer ratio and the adopted cut (dashed) sits on that structure would retain (marginal, not cascaded), over the full ra
Nota: POCO CHIARA ? INESATTA? SBAGLIATA?

---

194. [Evidenziato] Pagina 27
Testo: pted cut (dashed) sits on (marginal, not cascaded),
Nota: CHIARIRE

---

195. [Evidenziato] Pagina 28
Testo: by ng filter [11] sol least squares smoothed po
Nota: SII PIù specifioc. o qui. o in footnote. o in un'appendice. vedi tu

---

196. [Evidenziato] Pagina 28
Testo: ts or (d
Nota: and

---

197. [Evidenziato] Pagina 28
Testo: s are scaled by the flight's own !t (Sec. 2.7.6), so t and the window never crosses a segment boundary Sec. 2.7.6), the filter sees a uniform time base with
Nota: ok quindi quando ci si avvicina ad una segment boundary cosa succese? cosa succese alla window, diventa asimmetrica?

---

198. [Evidenziato] Pagina 28
Testo: ince heavily reconstructed flights but no bias over gaps this short.
Nota: poco chiaro questo pezzo.

---

199. [Evidenziato] Pagina 28
Testo: dynamica φc = 1/fc:
Nota: qundi x,y,z, avranno time scale diversi?

---

200. [Evidenziato] Pagina 28
Testo: fitted polynom p = 3 works
Nota: giustifichiamo l'uso di p=3 perchè è l'oridne più basso che funziona.

---

201. [Barrato] Pagina 28
Testo: must be at least quadratic, while a low order the result is insensitive to p within {2, 3, 4}.
Nota: —

---

202. [Evidenziato] Pagina 28
Testo: so the physical smoothing oor at = 5, the smallest Set the window per flight from the timescale, w = scale is the same across the fleet regardless of rat window a cubic fit accepts. For the few flights wi
Nota: frase poco chiara

---

203. [Evidenziato] Pagina 28
Testo: odd e – w
Nota: non si capisce chi è e cosa fa questa funzione "odd"

---

204. [Evidenziato] Pagina 28
Testo: noise band lies beyond those flights are in any Nyquist and the floor smooths slightly more than the noise sca case stratified out of the !t-sensitive observables (Sec. 2.7.6).
Nota: frase poco chiara

---

205. [Evidenziato] Pagina 28
Testo: arometric chan a longer one. Hz (= 5 s),
Nota: longer one rispetto a cosa

---

206. [Evidenziato] Pagina 28
Testo: barometric–GNSS asymmetry shows in The per-channel conditioning is kept indows are currently equal.
Nota: cosa intendi???

---

207. [Evidenziato] Pagina 28
Testo: The choice is validated on a held-out sample: the residual (raw minus smoothed) spectrum should be flat above, confirming that noise rather than signal was removed; the acceleration The choice is validated on a held-out sample: the residual (raw minus smoothed) spectrum should be flat above fc, confirming that noise rather than signal was removed; the acceleration distribution should stay within physical bounds; and the results should be stable under a should be flat above fc, confirming that noise rather than signal was removed; the acceleration distribution should stay within physical bounds; and the results should be stable under a one-step change of (implementation details: Sec. 2.6.7). distribution should stay within physical bounds; and the one-step change of w (implementation details: Sec. 2.6.7).
Nota: non chiaro. riscrivi

---

208. [Sottolineato] Pagina 28
Testo: position rajectory t is the e
Nota: ho notato che in giro per la tesi spesso si usa la aprola position ma seocndo me quando viene usata ci si riferisce solo all'horizonatl position. ecco, bene. voglio che ogni votla che ci si riferisce all'horizontal position lo si dica espliciemtnen e non is usi il temrine position al suo posto. quando si indica la poszion totlae , ointendo quela 3d, allora va specificato che is sta intendendo la position 3d.

---

209. [Sottolineato] Pagina 28
Testo: segment created by a split at a long gap, Sec. 2.7.6, keeps its parent flight's origin and The vertical is not re-zeroed: the take-o! height is retained, so the analysis keeps both rement that drives the dynamics (invariant to any o!set) and the absolute altitude for clock). The vertical is not re-zeroed: the take-o! height is retained, so the analysis keeps both the increment that drives the dynamics (invariant to any o!set) and the absolute altitude for interpretation (potential energy,

convective-boundary-layer structure). The horizontal velocity the increment that drives the dynamics (invariant to any o!set) and t interpretation (potential energy, convective-boundary-layer structure). is v() = ˙r(), with magnitude the horizontal speed =v; the ve
Nota: bene. allora qunado si parla dell'introduzione del sistema locale ENU va fatto capire che l'altituidne del punto izniale, del takeoff, non vinee posta a zero ma viene mantenuta quella misurata nell'istante che consideriamo di partenza della fase in volo.

---

210. [Evidenziato] Pagina 29
Testo: Preliminary characterization
Nota: il punto di questa sezione è anlaizzare le statistiche del dataset e delle sue osservabili principali in rifeirmento al dataste pultio, filtrato, interpolat. insomma il dataset che verrà effettivamente uato. ecco perchè, ad esempio, è di nuovo presente la stiatistica del native sampling itnerval. perchè questa è la statistica fatta solo sui voli che verranno effettivaemnte usati enon su tutti i voli del dataset come fatto in figura 2.6 e 2.7

---

211. [Evidenziato] Pagina 29
Testo: t of lightweight checks that validate and characterize the dataset at the ntation and are conceptually simpler basic assumptions, reveal pot level of individual trajectories. than the transport observable
Nota: non chiaro cosa vuol dire questa frase.

---

212. [Evidenziato] Pagina 29
Testo: ration Their ataset T, total horizontal path length, number of valid fixes and native sampling interval !t. Their ensemble distributions (the filtering-relevant ones in Fig. 2.3) characterise what the dataset contains, flag outliers, and, through !, set the per-flight smoothing window (Sec. 2.7.7). In ensemble distributions contains, flag outliers, the present data dura
Nota: frase poco chiara

---

213. [Evidenziato] Pagina 29
Testo: aset In the contains, flag outliers, and, through !t, set the per-flight smoothing window (Sec. 2.7.7). In the present data durations centre on a few hours and track lengths on ↓104 fixes, so the flight-level cuts fall in an essentially empty tail. the present data durations centre on a few ho flight-level cuts fall in an essentially empty tail
Nota: non cpiasco l'utilità di questa frse ne come fa a stare qui dato xhe non abbiamo ancora il dataset pulit.

---

214. [Evidenziato] Pagina 29
Testo: The scaling of the 1D marginal propagator (Section 3.1) requires the pic. Competition flights have preferred directions (race goal, orography, Isotropy check. The process to be isotropic. prevailing winds), so iso
Nota: va spiegato meglio. il punto è che se il processo è isotropo allora invece che studiar eil propagatore 2d ansiamo a studiar eil propagatore 1d.

215. [Evidenziato] Pagina 29
Testo: scaling of th Competition tropy must b
Nota: questi voli non sono competition nel senso di gara in contemporanea. sono solo flights che poi vanno a finire in una classifica. però in futuro è possinile che debba analizzare dei voli in competiazione contemporanea. duqneu sono due situazioni diverse. nella prima ci si può apsettare maggiormente una sorta di isotropia, mentre nel seocndo caso più no che si. inoltre per isotropia credo che ci se la possa aspettare solo a livello di ensamble e nona. alivello di sognolo volo, che ha una direzione ben specifica.

---

216. [Evidenziato] Pagina 29
Testo: s must be checked for statistical compatibility. or each stratum (e.g. a single wing class, or the flights of a mponent displacement variance()2as a function of the A lightwei single season) elapsed time
Nota: l'interesse è rivolto magiormanete verso le sottogategorie date dal tipo di ala usata. sempre all'interno di para o di hang. non ci sarà mai nessun pooled ensemble che mischierà hang e para.

---

217. [Evidenziato] Pagina 29
Testo: wing clas →E(t)2↑ de, the
Nota: perchè mai vado a vedere E(t) e non MSD(t). non lo trovo molto sensato

---

218. [Evidenziato] Pagina 29
Testo: A map of take-o! coordinates and displacement end-points ustering (France has distinct orographic zones: Alps, Pyrenee
Nota: —

---

219. [Evidenziato] Pagina 29
Testo: c clustering (France has distinct orographic zones: Alps, Pyrenees, Massif If flights from mountainous terrain di!er markedly from flatland flights, cation may be needed alongside temporal and aeronautical stratification. Central, plains). If flights from mountainous terrain di!er markedly from flatland flight geographic stratification may be needed alongside temporal and aeronautical stratification.
Nota: questa è un osservazione che va messa in compatibilty across strata oppure dici che è il casodi lasciarla qui?

---

220. [Sottolineato] Pagina 47
Testo: PSD sample (Figure 2.1a–c). A seeded ra larger than a spectral shape alone would need is visibly under-averaged (ragged) at a few hu
Nota: Sentence with an unclear meaning

---

221. [Sottolineato] Pagina 48
Testo: axis; flights with fewer than since at least one full segment is leaves 1,641 paragliders but Nseg = 25 is needed. 727 hang
Nota: What do you mean by this sentece? What does “a full segment” mean?

---

222. [Sottolineato] Pagina 48
Testo: z(t) is linearly interpolated onto a grid without introducing frequency content t0, t0 + !t, t0 beyond fNyq.
Nota: What do you mean?

---

223. [Evidenziato] Pagina 53
Testo: port estimators – the displaceme over a grid of cleaning strengths y stop moving; movement unde
Nota: It not clear what these “clear strenghts” really are
